Subject- PS26-Interdisciplinary Elective: AI, Power & Responsibility - Governing Intel

Final Project Topic Title - (Project No.1) Accuracy versus Sustainability: A Carbon-Aware Comparison of Machine Learning Models

Workflow structure to complete the project(Beginning to End):

Step 1: I am trying to work on a project, whose details are given below, for the module "Interdisciplinary Elective: AI, Power & Responsibility – Governing Intelligent Systems for Sustainability". I want to identify and finalize 5 interesting/real world research questions according to the current situation in Germany that fall in the same domain as this topic and a typical IEEE conference. More info on this topic:

“1. Carbon-Aware Machine Learning Model Selection

Title: Accuracy versus Sustainability: A Carbon-Aware Comparison of Machine Learning Models

Fundamental idea: Students compare different ML models not only by accuracy, but also by computational cost, runtime, model size, and energy/carbon proxy indicators.

Example 1: Compare Logistic Regression, Random Forest, XGBoost, and Neural Network on an energy demand dataset using accuracy, F1-score, runtime, and memory usage.

Example 2: Train multiple classifiers on a climate-risk dataset and create a ranking that balances predictive performance with low computational cost.”

Step 2: With these research questions, and the idea in place, identify publicly available dataset suitable for these research questions. Then follow up with the following steps:

Steps:

- Download one of the best suitable dataset especially that aligns with Germany.
- Explore the dataset. Name the raw dataset ‘data.csv’ and the processed data ‘preprocessed_content.csv’.

Step 3: For each research question, I want a separate ipynb file. It should save all figures as pdfs and all tables as csvs that take the raw data as input and give us the actual resulting tables and figures from the input. Generate the complete step by step methodology followed starting from the dataset to getting the results for each research question(figure and table). The methodology should follow this structure:
‘We select ABC dataset and EDF dataset . For each research question, please identify a publication ready figure and a publication ready table for depicting the answer for the research question. 5 publication-ready figures corresponding to these five RQs.
’

Generate the five ipynb files that take the raw data as input and give us the actual resulting tables and figures from the input, by implementing the above methodology.

Sample workflow:

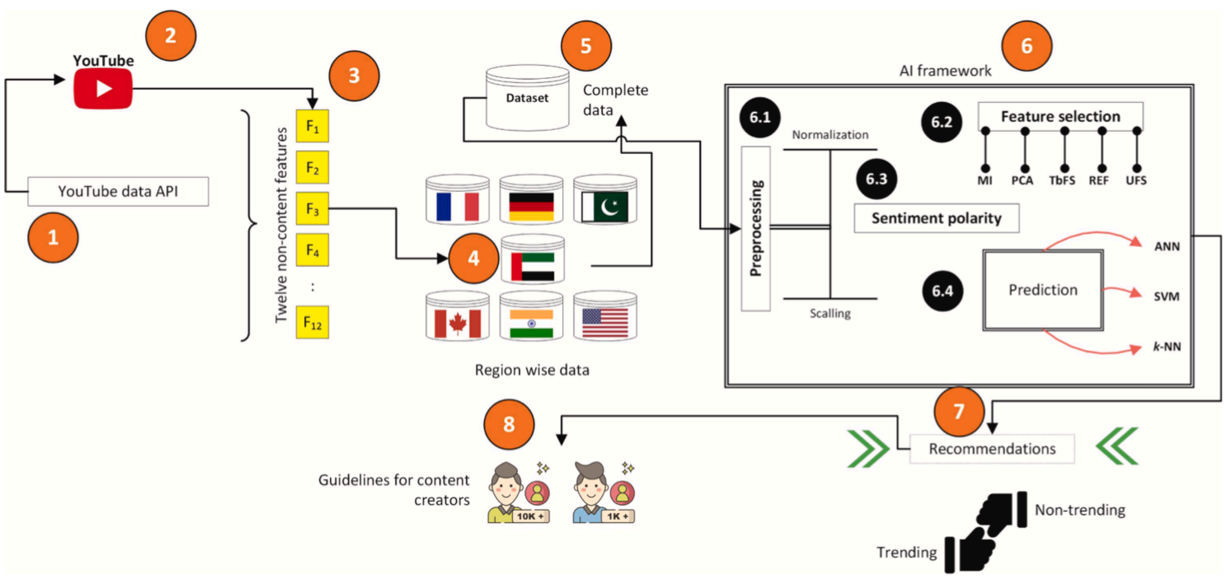


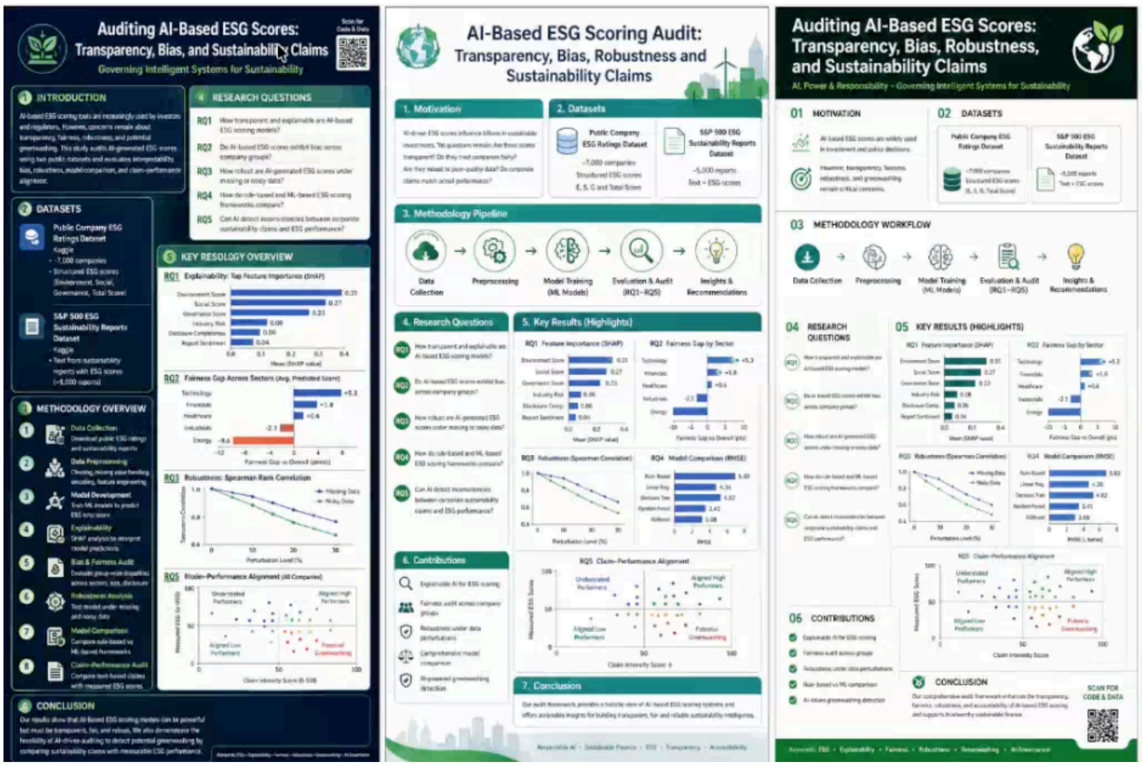
Fig. 1. The overall working of the proposed solution.

Step 4: Then, generate a scientific publication ready step by step workflow figure for this topic that is illustrative of the whole process, uses colors and numbers each step, in the same way as the attached workflow figure.

Step 5: Look at the generated image carefully, and identify the icons used in this image. Then generate the icons separately as a vector graphics file or as icons and save under the icons folder for later use just in case we want to edit the workflow figure in the ppt.

Step 6: Show me some sample scientific conference posters for this idea that I can use to present my work. (This poster should be later editable in powerpoint.)

Here are 3 sample posters to choose from are below (Note- this poster might not be clear. I took a screen shot, so make the image clear before):



At last, write a publication ready technical report with the publication ready workflow figure of the project along with all the figures, tables according to the RQs, ready for IEEE.

Here is the IEEE web link for the technical reports and white papers of what information should they contain and what they should look like-
https://ieee-pes.org/technical-activities/technical-reports-white-papers/
Scan a few technical reports and analyze the patterns before you write the technical report and before you generate workflow figures and data visualization figures.

I also have uploaded the IEEE technical report sample for your reference. I have also uploaded the exact workflow reference diagram.

I want the project and the report to be at least of a phd level(excellent). So, work according to that.

If you think there is something that doesn’t make sense from the steps/procedure that I gave you to complete this project, do tell me.